

Exploratory Data Analysis (EDA) Findings & Insights

1. Dataset Overview

- The dataset contains **891 passenger records** and **12 features**.
- It includes both **numerical** and **categorical** variables.
- The target variable is **Survived**, indicating whether a passenger survived the disaster.

2. Missing Values

Findings

- **Age** had **177 missing values**.
- **Cabin** had **687 missing values** (over 75% of the dataset).
- **Embarked** had **2 missing values**.

Action Taken

- Filled missing **Age** values with the **median**.
- Filled missing **Embarked** values with the **mode**.
- Dropped the **Cabin** column due to the large proportion of missing values.

Insight

Proper handling of missing values improves data quality and ensures reliable analysis.

3. Passenger Class Distribution

Findings

- Most passengers belonged to **Third Class (Pclass = 3)**.
- First and Second Class passengers were fewer.

Insight

The Titanic primarily carried third-class passengers, indicating that the majority of travelers were from the lower ticket category.

4. Survival Distribution

Findings

- Approximately **62%** of passengers did not survive.
- Approximately **38%** survived.

Insight

The survival rate was relatively low, showing that more passengers lost their lives than survived.

5. Age Distribution

Findings

- Most passengers were between **20 and 40 years old**.
- Very few passengers were above **60 years**.

Insight

The passenger population mainly consisted of young adults.

6. Fare Distribution

Findings

- Most ticket fares were relatively low.
- A small number of passengers paid exceptionally high fares.
- The distribution is highly right-skewed.

Insight

Ticket fares varied significantly, with a few premium-class passengers paying much higher prices.

7. Family Size (SibSp & Parch)

Findings

- Most passengers traveled alone.
- Only a small number traveled with siblings, spouses, parents, or children.

Insight

Solo travel was more common than family travel on the Titanic.

8. Outlier Analysis

Findings

The boxplots revealed:

- Several outliers in **Fare**.
- Some outliers in **Age**.

Insight

The Fare feature contains extreme values due to expensive first-class tickets.
These outliers are genuine observations and should not necessarily be removed.

9. Correlation Analysis (Heatmap)

Findings

- No pair of variables showed a very strong correlation.
- Fare showed a moderate relationship with Passenger Class.
- PassengerId had no meaningful relationship with other variables.

Insight

Most numerical features are only weakly correlated, suggesting that multiple factors influence passenger survival.

10. Pairplot Analysis

Findings

- No strong linear relationships among most numerical variables.
- First-class passengers generally paid higher fares.
- Most passengers had no siblings/spouses or parents/children aboard.
- Age distribution overlaps for both survivors and non-survivors.

Insight

Survival cannot be explained by a single numerical variable alone. Multiple factors contribute to passenger outcomes.

11. Gender Distribution *(if you created a countplot)*

Findings

- There were more male passengers than female passengers.

Insight

The passenger population was predominantly male.

12. Survival by Gender

Findings

- Female passengers had a much higher survival rate.
- Most male passengers did not survive.

Insight

Gender played a significant role in survival, consistent with the evacuation practice of prioritizing women and children.

13. Survival by Passenger Class

Findings

- First-class passengers had the highest survival rate.
- Third-class passengers had the lowest survival rate.

Insight

Passenger class significantly influenced survival chances, with higher-class passengers having better outcomes.

14. Distribution Shape

Findings

- **Age** is slightly right-skewed.
- **Fare** is heavily right-skewed.
- **SibSp** and **Parch** are concentrated at zero.

Insight

Most passengers traveled alone and paid lower fares, while only a few had very high ticket prices.